

Proof-of-Concept Report

Name: Taylor Myers

Date: 9/18/26



Table of Contents

Abstract.....	3
Introduction	3
Statement of Need.....	3
Assumptions.....	4
Description of Current Infrastructure	4
Cloud Service Providers	5
Project Details	6
Building a VPC on AWS.....	6
Challenges Encountered	7
Conclusion.....	7
References	8
Appendix - Screenshots	9

Abstract

This report presents a proof of concept for migrating part of Don & Associate's current IT infrastructure to Amazon Web Services (AWS). Don & Associates is a financial consulting company that currently relies on on-premises servers and local network resources but is preparing for future growth and expansion. The proof of concept demonstrates how AWS can support the company by creating a Virtual Private Cloud (VPC), configuring subnets and security groups, and launching an EC2 instance to host a web server. The report also reviews major cloud service providers, explains the steps used to build the AWS environment, discusses challenges encountered during the lab, and evaluates how the solution could support the company's future business needs.

Introduction

Don & Associates is a financial consulting company that is preparing to expand into neighboring states in the Northeast. The company currently relies on on-premises servers and local network resources to support its business operations. As the company grows, it will need to support more employees, clients, and locations while also increasing its computing and storage resources.

Amazon Web Services (AWS) was selected as the cloud provider for Don & Associates because of its scalability, security, broad range of services, and support resources (Panagou et al., 2025). This proof-of-concept project demonstrates how AWS can support the company by creating a Virtual Private Cloud (VPC), configuring networking and security settings, and launching a web server in the cloud.

Statement of Need

Don & Associates currently relies on traditional on-premises infrastructure, which can become more difficult and expensive to maintain as the company grows. Expanding into additional states will increase the number of employees, clients, and locations that depend on company systems. This will also increase the need for computing power, storage, network resources, and secure access from multiple locations.



Moving to the cloud can help Don & Associates reduce its dependence on physical hardware and make it easier to scale resources as business needs change. A cloud environment can also provide employees with more flexible access to company systems while supporting future growth without requiring the company to constantly purchase and install additional on-premises equipment.

Assumptions

For this proof of concept, it is assumed that Don & Associates will continue expanding into additional locations and will need more flexible computing and network resources. It is also assumed that the company wants to reduce its dependence on physical on-premises infrastructure while maintaining secure and reliable access to its systems. The AWS environment created in this project is intended only as a demonstration and does not represent a complete production environment. Additional security, monitoring, backup, and configuration controls would be needed before the company could use the environment for actual business operations.

Description of Current Infrastructure

Don & Associates currently uses on-premises servers and local network resources to support its daily business operations. Company applications and data are hosted internally instead of through a public cloud provider, so the company is responsible for managing its own hardware, storage, maintenance, and system availability. Employees also rely on the existing network environment to access company systems and files.

As the company grows, this infrastructure may become more difficult to manage. Increasing storage or computing capacity can require purchasing and installing additional physical equipment, and hardware failures or local outages could affect access to company systems. Supporting multiple office locations can also make centralized access and management more challenging.



Cloud Service Providers

The three major cloud service providers considered for Don & Associates are Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). Each provider offers cloud-based computing, networking, security, and application services that can help businesses reduce their dependence on physical infrastructure (Olariu et al., 2024).

Amazon Web Services offers SaaS, PaaS, and IaaS options along with a broad range of cloud services (Panagou et al., 2025). AWS provides strong scalability, security, reliability, and data protection (Panagou et al., 2025). It also offers extensive documentation, training, and support resources (Panagou et al., 2025). One disadvantage is that pricing can become complex because costs vary based on the services used and the amount of resources consumed (Panagou et al., 2025).

Microsoft Azure also offers SaaS, PaaS, and IaaS services (Ivan et al., 2024). Its strengths include integration with Microsoft products, strong security features, high availability, and flexible billing options (Ivan et al., 2024). Azure can be especially useful for organizations that already rely heavily on Microsoft technologies (Ivan et al., 2024).

Google Cloud Platform provides IaaS, PaaS, and serverless cloud services (Olariu et al., 2024). It offers computing, storage, analytics, machine learning, and application development tools (Olariu et al., 2024). GCP is particularly strong in analytics, artificial intelligence, and developer-focused services (Olariu et al., 2024). One limitation is that it may not be the best fit for every organization depending on existing systems and business needs (Olariu et al., 2024).

For Don & Associates, AWS was selected because of its broad service selection, scalability, security, and support resources align well with the company's expected growth and infrastructure needs (Panagou et al., 2025).



Project Details

The proof-of-concept project was completed using Amazon Web Services. The main goal was to create a cloud environment that could demonstrate how Don & Associates could begin moving part of its existing infrastructure to AWS. The project involved creating a Virtual Private Cloud (VPC), adding subnets, configuring a security group, and launching an EC2 instance to operate as a web server. These steps created a basic cloud network environment that could be used to demonstrate how company resources can be hosted and accessed through AWS.

Building a VPC on AWS

The first step in the proof of concept was to create a Virtual Private Cloud in AWS. The VPC provides an isolated network environment where Don & Associates can host cloud resources. After creating the VPC, additional subnets were added to organize resources within the network and support communication between various parts of the environment. A security group was then created to control the traffic allowed to reach the resources inside the VPC. These settings helped establish the basic network structure and security controls needed before launching the web server.

Launching a Web Server on AWS

After the VPC and security settings were configured, an Amazon EC2 instance was launched to act as the web server for proof of concept. The instance was placed inside the VPC so it could use the network and security configuration that had already been created. The web server was then configured and customized so that it could host a basic webpage. Once the instance was running, the web server was tested to confirm that it could be accessed successfully. This demonstrated how Don & Associates could host web-based resources in AWS instead of relying only on its current on-premises infrastructure.



Challenges Encountered

One challenge during the proof-of-concept process was making sure each AWS component was configured correctly before moving to the next step. The VPC, subnets, security group, and EC2 instance all had to work together for the web server to function properly. It was also important to verify that the correct network and security settings were selected when launching the instance. Following the lab instructions carefully and checking each configuration helped complete the environment successfully.

Conclusion

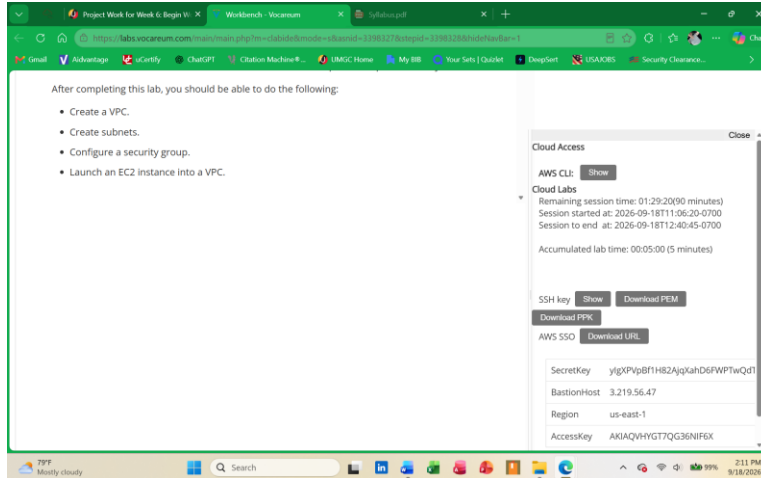
The proof of concept demonstrated how Don & Associates could begin using Amazon Web Services to support its future cloud environment. By creating a VPC, configuring subnets and security settings, and launching an EC2 instance as a web server, the project showed how AWS can provide a flexible and scalable alternative to the company's current on-premises infrastructure. This solution could help Don & Associates reduce its dependence on physical hardware while supporting future growth, additional locations, and increasing technology needs. The proof of concept also provides a starting point for evaluating how a larger migration to AWS could be planned and implemented (Panagou et al., 2025).

References

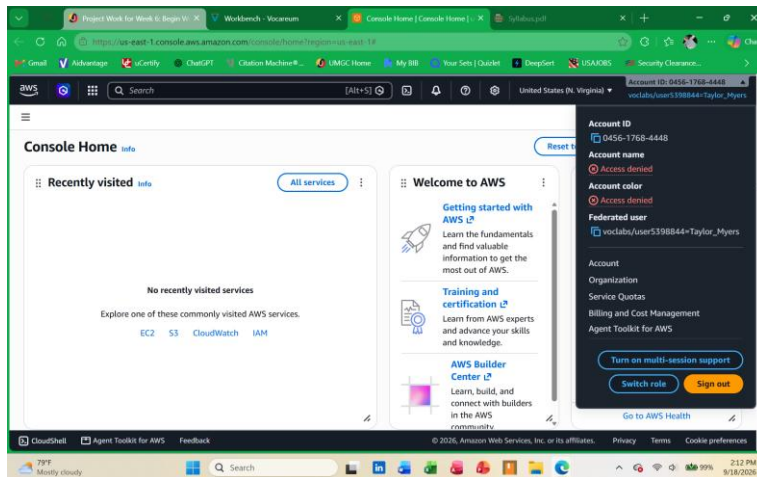
- Ivan, Sergiu C., Győrödi, Robert Ş., & Győrödi, Cornelia A. (2024). Sentiment Analysis Using Amazon Web Services and Microsoft Azure. *Big Data and Cognitive Computing*, 8(12), 166. <https://doi.org/10.3390/bdcc8120166>
- Olariu, Florin, Alboaie, Lenuța, & Grănescu, Rareș. (2024). Beyond Cloud Boundaries: An Analytical Case Study on the Migration of a Modular Monolith to Azure, AWS, and Google Cloud Platform. *Procedia Computer Science*, 246, 2782–2791. <https://doi.org/10.1016/j.procs.2024.09.393>
- Panagou, Ioannis Chrysovalantis, Katsoulis, Stylianos, Nannos, Evangelos, Zantalis, Fotios, & Koulouras, Grigorios. (2025). A Comprehensive Evaluation of IoT Cloud Platforms: A Feature-Driven Review with a Decision-Making Tool. *Sensors (Basel, Switzerland)*, 25(16), 5124. <https://doi.org/10.3390/s25165124>

Appendix - Screenshots

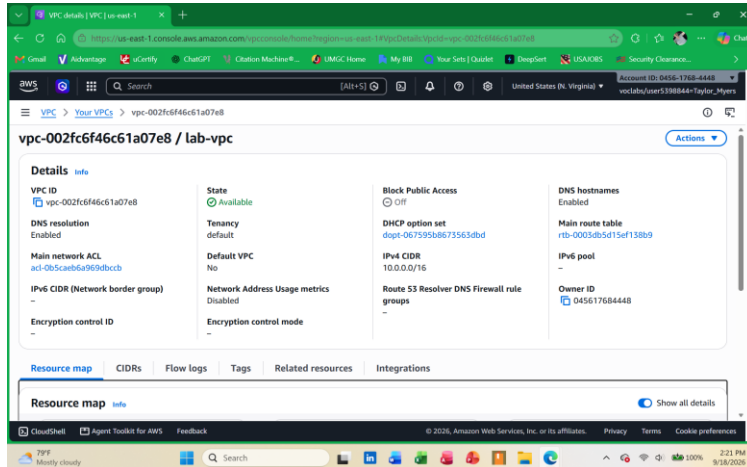
Start Lab



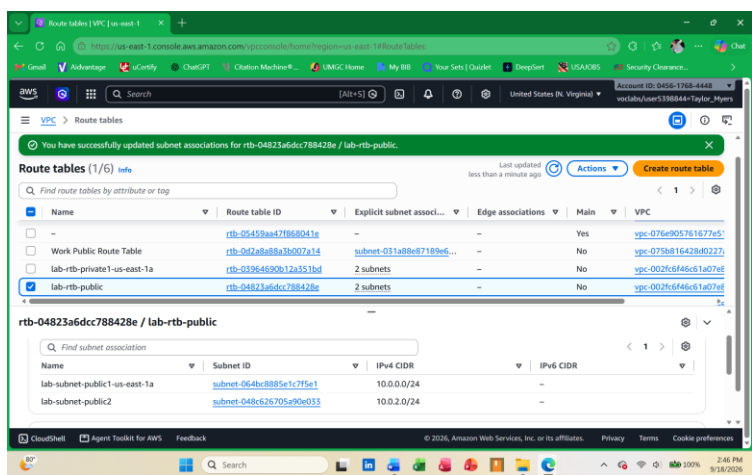
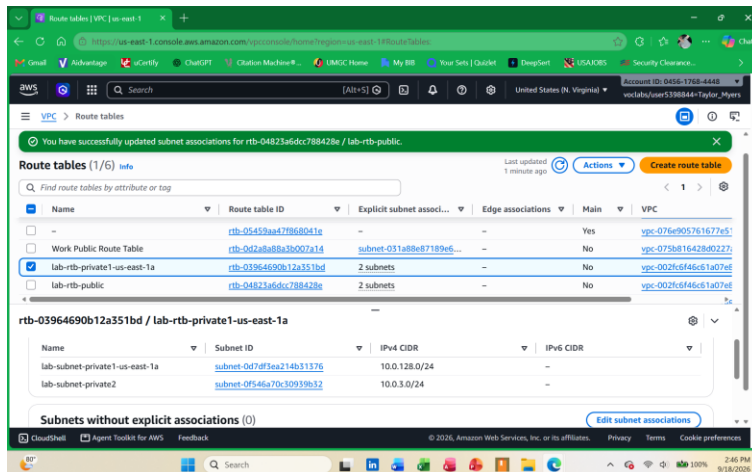
AWS Management Console Name



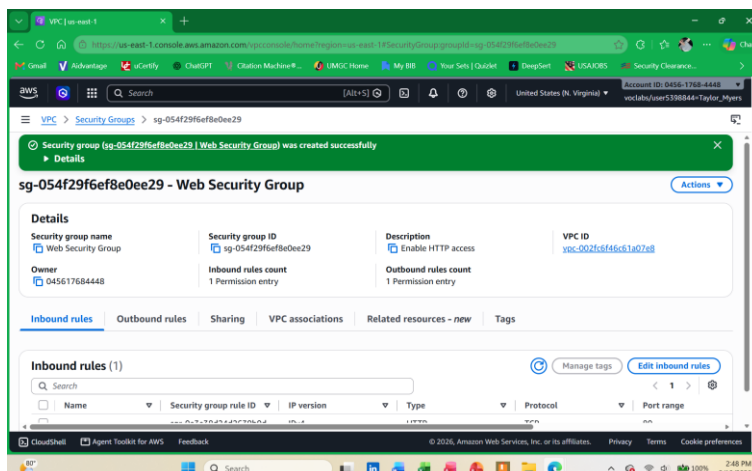
Task 1: Create Your VPC



Task 2: Create Additional Subnets



Task 3: Create a VPC Security Group



Task 4: Launch a Web Server Instance

Instances [EC2] us-east-1

Instances (1/2) info

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS
Bastion Host	i-01f3f3490c910edcd	Running	t2.micro	2/2 checks passed	us-east-1a	ec2-3-219-56-47.comp...
Web Server 1	i-00d2e26a2112d0947	Running	t2.micro	2/2 checks passed	us-east-1b	ec2-3-86-225-134.com...

i-00d2e26a2112d0947 (Web Server 1)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary info

Instance ID: i-00d2e26a2112d0947

Public IPv4 address: 3.86.225.134 | open address

Private IPv4 addresses: 10.0.2.28

Public DNS: ec2-3-86-225-134.com...

Instance state: Running

aws Load Test RDS

Meta-Data

Meta-Data	Value
InstanceId	i-00d2e26a2112d0947
Availability Zone	us-east-1b

Current CPU Load: 0%

